

TECHNICAL DESIGN: MULTI-SOURCE CANDIDATE DATA TRANSFORMER

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1. PIPELINE ARCHITECTURE & DATA FLOW

Ingestion Read CSV, JSON, TXT, PDF from CLI inputs.	Parsing Extract candidate info using rules/regex.	Normalization Format phones (E.164), dates (YYYY-MM), countries.	Merging Deduplicate via keys & trust hierarchies.	Projection Reshape profile using custom runtime config.	Validation Validate fields & output final JSON.
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2. CANONICAL SCHEMA & NORMALIZATION STANDARDS

Field	Type / Target Format	Normalization Rules	Example Raw -> Target
candidate_id	string (UUID v5)	Deterministic namespace UUID generated from primary email.	john@doe.com -> fd8b0826-...
full_name	string	Title-cased name, punctuation stripped.	john d. -> John D.
emails / phones	string[]	Emails lowercase. Phones converted to E.164 standard.	+1 (555) 019-9234 -> +15550199234
location	{ city, region, country }	Country standardized to ISO-3166 alpha-2.	United States -> US
skills	[[name, confidence, sources]]	Mapped to standard skill names list. Confidence assigned.	reactjs -> React (Conf: 1.0)
experience	[[company, title, start, end]]	Merged history; dates normalized to YYYY-MM.	Jan 2020 -> 2020-01

3. MERGE POLICY & CONFIDENCE SCORING METHODOLOGY

Match Rules: Profiles are linked if they share a normalized email or phone. Name-only matches are resolved using a token set similarity score.

Trust Hierarchy: ATS JSON (1.0) > Recruiter CSV (0.9) > Unstructured Resume (0.8) > Recruiter Notes (0.6). Merged fields (e.g. name, location, job title) are selected from the most trusted source.

Overall Confidence Formula:	$Confidence\ Score = (Primary\ Source\ Trust * 0.5) + (Profile\ Completeness * 0.3) + (Multi-Source\ Agreement\ Bonus * 0.2)$	Agreement Bonus: +10% per extra agreeing source (max 20%). Completeness Ratio: Fraction of core fields present.
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4. CONFIGURABLE PROJECTION & EDGE CASES HANDLED

Runtime Configurable Projection Engine: Field Selection & Remapping: Reshapes canonical profile schema. Extracts path values like emails[0], skills[].name. Per-field Normalization: Allows overrides (e.g. E164 phone formatting or skill canonicalization). Missing Values Strategy: Evaluates global policy rules (null, omit, error) for absent fields. Provenance Control: Toggle switch to include/exclude metadata tracking and source trust confidence scores.	Edge Case Handling & Mitigations: Name-only Matches: Compares name tokens when email/phone are absent, preventing duplicates for matching candidates. Experience Overlaps: Deduplicates overlapping jobs (company + dates), merging details based on trust scores. Garbage Inputs: Parser errors degrade gracefully instead of crashing; values resolve to null, never invented. Required Field Failures: Throws clear validation errors if a required field is missing under on_missing: 'error'. Descoped: NLP name/entity models and skill taxonomy auto-updates (handled deterministically under time pressure).
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